Akhila Pisupati 225 HW3 Problem 2.8 2/2/26

$P_{+y} = |\langle +_y | +_n \rangle|^2$ $\theta = \frac{\pi}{4};\ \phi = \frac{5\pi}{3}$

$\langle +_y | = \frac{1}{\sqrt{2}}\langle + | - \frac{1}{\sqrt{2}} i \langle - |$ $|+_n\rangle = \cos(\frac{\theta}{2})|+\rangle + \sin(\frac{\theta}{2}) e^{i\phi}|-\rangle$

$= \cos(\frac{\pi}{8})|+\rangle + \sin(\frac{\pi}{8}) e^{i\frac{5\pi}{3}}|-\rangle$

$P_{+y} = \left|\left[\frac{1}{\sqrt{2}}\langle + | - \frac{1}{\sqrt{2}} i \langle - |\right]\left[\cos(\frac{\pi}{8})|+\rangle + \sin(\frac{\pi}{8}) e^{i\frac{5\pi}{3}}|-\rangle\right]\right|^2$

$= \left|\frac{1}{\sqrt{2}}\cos(\frac{\pi}{8}) - \frac{1}{\sqrt{2}} i \sin(\frac{\pi}{8}) e^{i\frac{5\pi}{3}}\right|^2 = \frac{1}{2}\left|\cos(\frac{\pi}{8}) - i\sin(\frac{\pi}{8})\left(\cos(\frac{5\pi}{3}) + i\sin(\frac{5\pi}{3})\right)\right|^2$

$e^{i\frac{5\pi}{3}} = \cos(\frac{5\pi}{3}) + i\sin(\frac{5\pi}{3})$

$= \frac{1}{2}\left|\cos(\frac{\pi}{8}) + \sin(\frac{\pi}{8})\sin(\frac{5\pi}{3}) - i\sin(\frac{\pi}{8})\cos(\frac{5\pi}{3})\right|^2$

$= \frac{1}{2}\left(\sqrt{\left(\cos(\frac{\pi}{8}) + \sin(\frac{\pi}{8})\sin(\frac{5\pi}{3})\right)^2 + \left(-\sin(\frac{\pi}{8})\cos(\frac{5\pi}{3})\right)^2}\right)^2$

$= \frac{1}{2}\left[\cos^2(\frac{\pi}{8}) + 2\sin(\frac{\pi}{8})\cos(\frac{\pi}{8})\sin(\frac{5\pi}{3}) + \sin^2(\frac{\pi}{8})\sin^2(\frac{5\pi}{3}) + \sin^2(\frac{\pi}{8})\cos^2(\frac{5\pi}{3})\right]$

$\approx .1938$

$P_{-y} = \left|\left[\frac{1}{\sqrt{2}}\langle + | + \frac{1}{\sqrt{2}} i \langle - |\right]\left[\cos(\frac{\pi}{8})|+\rangle + \sin(\frac{\pi}{8}) e^{i5\pi/3}|-\rangle\right]\right|^2$

$= \left|\frac{1}{\sqrt{2}}\cos(\frac{\pi}{8}) + \frac{1}{\sqrt{2}} i \sin(\frac{\pi}{8}) e^{i\frac{5\pi}{3}}\right|^2 = \frac{1}{2}\left|\cos(\frac{\pi}{8}) + i\left[\sin(\frac{\pi}{8})\left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}\right)\right]\right|^2$

$= \frac{1}{2}\left|\cos\frac{\pi}{8} + i\sin\frac{\pi}{8}\cos\frac{5\pi}{3} - \sin\frac{\pi}{8}\sin\frac{5\pi}{3}\right|^2 = \frac{1}{2}\left(\sin^2\frac{\pi}{8}\cos^2\frac{5\pi}{3} + \cos^2\frac{\pi}{8} - 2\cos\frac{\pi}{8}\sin\frac{\pi}{8}\sin\frac{5\pi}{3} + \sin^2\frac{\pi}{8}\sin^2\frac{5\pi}{3}\right)$

$\approx .8062$

$\boxed{\text{Problem 3.5}}$ Akhila Pisupati 2/3/26

(a) Measuring spin in the x-direction

Possible results: $S_x = +\frac{\hbar}{2}$ or $S_x = -\frac{\hbar}{2}$ because we are dealing with a spin-$\frac{1}{2}$ particle

Probabilities: Since $|\psi(t=0)\rangle = |+_z\rangle$, the probability of achieving either S_x result is 50% (see SG experiment 2)

(b) Finding state of system in S_z basis at time t

$|\psi(0)\rangle = |+\rangle$

The applied magnetic field is $\vec{B} = B_0 \hat{y}$ (in the y-direction)

so $H = \omega_0 S_y \doteq \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{i\hbar\omega_0}{2} \\ \frac{i\hbar\omega_0}{2} & 0 \end{pmatrix}$

$\det|H - \lambda I| = \begin{vmatrix} -\lambda & -i\hbar\omega_0/2 \\ \frac{i\hbar\omega_0}{2} & -\lambda \end{vmatrix} = \lambda^2 + i^2 \frac{\hbar^2\omega_0^2}{4} \rightarrow \lambda^2 = \left(\frac{\hbar\omega_0}{2}\right)^2$ so $\lambda = \pm\frac{\hbar\omega_0}{2}$

So, $\lambda_1 = \frac{\hbar\omega_0}{2}$ and $\lambda_2 = -\frac{\hbar\omega_0}{2}$

looked up online

$|\psi(0)\rangle$ in y-basis $= \left(|+_y\rangle\langle +_y| + |-_y\rangle\langle -_y|\right)|+\rangle = \frac{1}{\sqrt{2}}|+_y\rangle + \frac{1}{\sqrt{2}}|-_y\rangle$

So $|\psi(t)\rangle$ in y-basis $= \frac{1}{\sqrt{2}} e^{-i\omega_0 t/2}|+_y\rangle + \frac{1}{\sqrt{2}} e^{i\omega_0 t/2}|-_y\rangle$

Now back to z Basis:

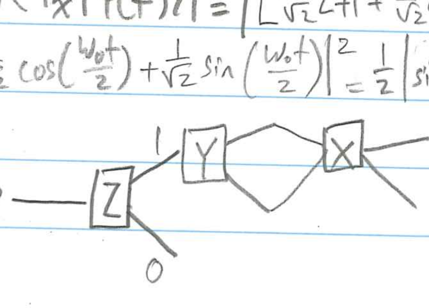
$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} e^{-i\omega_0 t/2}\left[\frac{1}{\sqrt{2}}|+\rangle + \frac{i}{\sqrt{2}}|-\rangle\right] + \frac{1}{\sqrt{2}} e^{i\omega_0 t/2}\left[\frac{1}{\sqrt{2}}|+\rangle - \frac{i}{\sqrt{2}}|-\rangle\right]$$

$$= \underbrace{\left(\frac{1}{2}e^{-i\omega_0 t/2} + \frac{1}{2}e^{i\omega_0 t/2}\right)}_{=\cos\left(\frac{\omega_0 t}{2}\right)\text{ (known identity)}}|+\rangle + \underbrace{\left(\frac{1}{2}ie^{-i\omega_0 t/2} - \frac{1}{2}ie^{i\omega_0 t/2}\right)}_{= i^2\left(\frac{e^{i\omega_0 t/2} - e^{-i\omega_0 t/2}}{2i}\right) = -i^2\sin\left(\frac{\omega_0 t}{2}\right) = \sin\left(\frac{\omega_0 t}{2}\right)}|-\rangle$$

$$|\psi(t)\rangle = \cos\left(\frac{\omega_0 t}{2}\right)|+\rangle + \sin\left(\frac{\omega_0 t}{2}\right)|-\rangle$$

(c) $P_{+x} = |\langle +_x|\psi(t)\rangle|^2 = \left|\left[\frac{1}{\sqrt{2}}\langle +| + \frac{1}{\sqrt{2}}\langle -|\right]\left[\cos\left(\frac{\omega_0 t}{2}\right)|+\rangle + \sin\left(\frac{\omega_0 t}{2}\right)|-\rangle\right]\right|^2$

$= \left|\frac{1}{\sqrt{2}}\cos\left(\frac{\omega_0 t}{2}\right) + \frac{1}{\sqrt{2}}\sin\left(\frac{\omega_0 t}{2}\right)\right|^2 = \frac{1}{2}\left|\sin\left(\frac{\omega_0 t}{2}\right) + \cos\left(\frac{\omega_0 t}{2}\right)\right|^2$

(d) $|\psi(0)\rangle$ 

$\boxed{\text{Problem 3.13}}$ Akhila Bisupati 2/3/26

(a) $H = \begin{pmatrix} E_0 & 0 & A \\ 0 & E_1 & 0 \\ A & 0 & E_0 \end{pmatrix}$ and basis states $|1\rangle, |2\rangle, |3\rangle$

$\det|H - \lambda I| = \begin{vmatrix} E_0-\lambda & 0 & A \\ 0 & E_1-\lambda & 0 \\ A & 0 & E_0-\lambda \end{vmatrix} = (E_0-\lambda)^2(E_1-\lambda) - A^2(E_1-\lambda) = 0$

$= (E_1-\lambda)[(E_0-\lambda)^2 - A^2] = 0$, so $\lambda_1 = E_1$ and $(E_0-\lambda)^2 = A^2$ so $E_0 - \lambda = \pm A$

$\lambda_2 = E_0 - A$ and $\lambda_3 = E_0 + A$

Not sure if relevant? (never mind!) $|n\rangle$

Assume $|1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, |2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, |3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ are the eigenvectors

$|\Psi(0)\rangle = |2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ so $|\Psi(t)\rangle = e^{-iE_2t/\hbar}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ e^{-iE_2t/\hbar} \\ 0 \end{pmatrix}$

$P_2 = |\langle 2|\Psi(t)\rangle|^2 = |\langle 2|[e^{-iE_2t/\hbar}|2\rangle]|^2 = |e^{-iE_2t/\hbar}|^2 = 1$

(b) $|\Psi(0)\rangle = |3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ so $|\Psi(t)\rangle = \begin{pmatrix} 0 \\ 0 \\ e^{-iE_3t/\hbar} \end{pmatrix}$

$P_3 = |\langle 3|\Psi(t)\rangle|^2 = |(0\ 0\ 1)e^{-iE_3t/\hbar}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}|^2 = |e^{-iE_3t/\hbar}|^2 = 1$??

I know the eigenvalues are relevant somehow but I can't figure out where/how... because $|3\rangle = |E_0 + A\rangle$ but where that comes into play I'm unsure...

$|\Psi(t)\rangle = \sum_n c_n e^{-iE_nt/\hbar}|E_n\rangle$

Problem 3.15   Akhilan Pisupati   2/3/26

Schroedinger equation: $i\hbar\frac{d}{dt}|\Psi(t)\rangle = H(t)|\Psi(t)\rangle$

$$i\hbar\frac{d}{dt}\left[\sum_n c_n e^{-iE_nt/\hbar}\right]|E_n\rangle = H(t)\sum_n c_n e^{-iE_nt/\hbar}|E_n\rangle$$

$$\frac{d}{dt}\left[e^{-iE_nt/\hbar}\right] = -\frac{iE_n}{\hbar}e^{-iE_nt/\hbar}$$

$$i\hbar\left(-\frac{i}{\hbar}E_n\right)\sum_n c_n e^{-iE_nt/\hbar}|E_n\rangle = -i^2E_n\sum_n c_n e^{-iE_nt/\hbar}|E_n\rangle = E_n\sum_n c_n e^{-iE_nt/\hbar}|E_n\rangle = H\sum_n c_n e^{-iE_nt/\hbar}|E_n\rangle$$

Dividing both sides by $\sum_n c_n e^{-iE_nt/\hbar}$, $E_n|E_n\rangle = H|E_n\rangle$

which is the energy eigenvalue equation. So this holds true (plus an accidental derivation)

Energy eigenvalue equation: But if we plug our superposition state into the energy eigenvalue equation directly, it doesn't work! ☹

$$H|\Psi(t)\rangle = H\sum_n c_n e^{-iE_nt/\hbar}|E_n\rangle = E|\Psi(t)\rangle = E\sum_n c_n e^{-iE_nt/\hbar}|E_n\rangle$$

(Because there's only 1 eigenvalue E) and there's no E that can satisfy this equation.
not E_n

TWO-STATE TIME DEPENDENCE

Name Akhila Basapak

QM
HW-1

Time dependence for stationary and non-stationary states

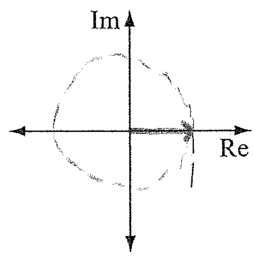
Suppose that the **z-component** of spin for an electron is measured just before $t = 0$ and is found to be $+\hbar/2$. A uniform magnetic field with magnitude B_0 is turned on in the **z-direction** at $t = 0$. $\vec{B} = B_0\hat{z}$

1. Write the state vector for this electron as a function of time in the form $\alpha(t)|+\rangle + \beta(t)|-\rangle$ (in the S_z basis). (*Hint:* What are $\alpha(t)$ and $\beta(t)$ at time $t = 0$?)

   $|\Psi(0)\rangle = |+\rangle = \alpha(0)|+\rangle + \beta(0)|-\rangle$ so $\alpha(0) = 1$ and $\beta(0) = 1$

   $|\Psi(t)\rangle = e^{-i\frac{\omega_0}{2}t}|+\rangle$

2. Graph $\alpha(t)$ on the set of axes at right by following the procedure from part I of Worksheet 3.



3. For $t > 0$, what is the probability that a measurement of the **z-component** of spin for this electron results in $+\hbar/2$? Justify your answer using your graph.

   100% because the probability amplitude of $|\Psi(t)\rangle$ is $= |\alpha(t)| = |e^{-i\frac{\omega_0}{2}t}| = 1$ (since $|e^{i\theta}| = 1$ for any arbitrary θ)
   (It is a "global phase")

4. For $t > 0$, what is the probability that a measurement of the **x-component** of spin for this electron results in $+\hbar/2$? Show your work. (Assume no measurements were made in the previous question.)

   $P_{+x} = |\langle +_x|\Psi(t)\rangle|^2 = |[\frac{1}{\sqrt{2}}\langle +| + \frac{1}{\sqrt{2}}\langle -|][e^{-i\frac{\omega_0}{2}t}|+\rangle]|^2$

   $\langle +_x| = \frac{1}{\sqrt{2}}\langle +| + \frac{1}{\sqrt{2}}\langle -|$ $\quad = |\frac{1}{\sqrt{2}} e^{-i\frac{\omega_0}{2}t}|^2 = \frac{1}{2}$

5. Which of the probabilities you computed in questions 3 and 4, if either of them, depend on time?

   Neither are time dependent

The term *stationary state* may be used to describe the state of the electron in questions 1 through 5.

6. The state vector for the electron you wrote down in question 1 should be dependent on time. Explain how this is consistent with the fact that the electron is in a stationary state.

   There is only one eigenvector here and so there's only one change in phase (whose magnitude is 1) → any measurement is stationary
   If there was >1 phase (/basis vector) any measurement out of these bases would be time dependent (there would be a "net phase" present ≠ 0) which means the state wouldn't be stationary

7. Consider the student discussion below.

Student 1: "The probability that a measurement of the z-component of spin resulted in $+\hbar/2$ was independent of time for both the state in questions 1 through 5 of this homework and for the state in part I, section II of Worksheet 3, so they are both stationary states."

Student 2: "I disagree. The state that started as spin up in the z-direction is stationary because *all* probabilities are independent of time, even though the state vector does depend on time."

Student 3: "I think it depends on what we measure. Any state is stationary if we measure the spin in the z-direction, but not stationary if we were measure the spin in the x-direction."

With which student(s) do you agree, if any? For each student who is incorrect, identify the flaw(s) in that student's reasoning. Explain.

Student 1 is wrong: in worksheet 3, $|\chi\rangle = \alpha_0 e^{-i\omega_0 t/2}|+\rangle + \beta_0 e^{+i\omega_0 t/2}|-\rangle$ is a superposition of multiple basis vectors so it cannot be stationary, since there are different phases ($e^{-i\omega_0 t/2}$ and $e^{+i\omega_0 t/2}$)

Student 2 is correct: the probabilities of the spin-up-z state are 0 and 1, which are obviously time independent, but $|\psi(t)\rangle = e^{-i\frac{\omega_0}{2}t}|+\rangle$ which obviously is dependent due to the phase.

Student 3 is wrong: Not "any state"; the state is only stationary if its time dependence is only in the global phase ($e^{-i\omega t}$ or some related form). The direction by itself makes no difference; you could easily change the state's basis and it wouldn't suddenly become stationary or not.

Suppose that the **z-component** of spin for a different electron is measured just before $t = 0$ and is found to be $+\hbar/2$ (same as with the first electron). A uniform magnetic field with magnitude B_0 is turned on in the **x-direction** at $t = 0$ (different than with the first electron).

8. Express the state vector for this electron as a function of time in the form $\alpha(t)|+\rangle + \beta(t)|-\rangle$ (in the S_z basis). Show your work. (*Hint:* First determine $\alpha(t = 0)$ and $\beta(t = 0)$, then perform an appropriate change of basis to determine the time-dependence before changing back to the S_z basis.)

$\alpha(0)=1$ and $\beta(0)=0$ so $|\psi(0)\rangle = |+\rangle = \frac{1}{\sqrt{2}}|+_x\rangle + \frac{1}{\sqrt{2}}|-_x\rangle$

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}e^{-i\frac{\omega_0}{2}t}|+_x\rangle + \frac{1}{\sqrt{2}}e^{+i\frac{\omega_0}{2}t}|-_x\rangle$$

In the z basis, $|\psi(t)\rangle = \frac{1}{\sqrt{2}}e^{-i\frac{\omega_0 t}{2}}\left(\frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle\right) + \frac{1}{\sqrt{2}}e^{+i\frac{\omega_0}{2}t}\left(\frac{1}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle\right)$

$$= \frac{1}{2}e^{-i\frac{\omega_0 t}{2}}|+\rangle + \frac{1}{2}e^{-i\frac{\omega_0 t}{2}}|-\rangle + \frac{1}{2}e^{i\omega_0 t/2}|+\rangle - \frac{1}{2}e^{i\omega_0 t/2}|-\rangle$$

$$= \left(\frac{1}{2}e^{-i\omega_0 t/2} + \frac{1}{2}e^{i\omega_0 t/2}\right)|+\rangle + \left(\frac{1}{2}e^{-i\omega_0 t/2} - \frac{1}{2}e^{i\omega_0 t/2}\right)|-\rangle$$

$$\cos\left(\frac{\omega_0 t}{2}\right) = \frac{e^{i\left(\frac{\omega_0 t}{2}\right)} + e^{-i\left(\frac{\omega_0 t}{2}\right)}}{2}, \quad \sin\left(\frac{\omega_0 t}{2}\right) = \frac{e^{i\left(\frac{\omega_0 t}{2}\right)} - e^{-i\left(\frac{\omega_0 t}{2}\right)}}{2i} \Rightarrow + \frac{e^{-i\frac{\omega_0 t}{2}} - e^{i\frac{\omega_0 t}{2}}}{2i} = -\sin\left(\frac{\omega_0 t}{2}\right)$$

$$|\psi(t)\rangle = \cos\left(\frac{\omega_0 t}{2}\right)|+\rangle - i\sin\left(\frac{\omega_0 t}{2}\right)|-\rangle$$

9. Is the electron described by the state vector in question 8 in a stationary state? Explain.

No, because the probability amplitudes are not only time dependent but they change at different rates, so there exists two independent phases which don't cancel out. Thus the "net global phase" of the state depends on time

10. Based on your work in this homework, can an electron be in a stationary state but *not* in an energy eigenstate? Can an electron be in an energy eigenstate but *not* in a stationary state? Explain.

No for both:

Stationary but not in an energy eigenstate: If stationary, the probabilities must be constant over time but that's also a condition of energy eigenstates because if your state isn't an eigenstate, there's multiple energy eigenvalues so... I got stuck :(

Energy eigenstate but not stationary — to be in an energy eigenstate, $|\psi\rangle$ has to satisfy $H|\psi\rangle = E|\psi\rangle$. This means also that it must satisfy the schroedinger equation $i\hbar \frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle$, of which, as is laid out in the book, the solution is $|\psi(t)\rangle = e^{\frac{-iE_n t}{\hbar}}|\psi(t)\rangle$. This is actually just the definition of a stationary state! So, this cannot hold: if energy eigenstate, has to be stationary.